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Relation-aware graph structure embedding with co-contrastive learning for drug–drug interaction prediction

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ABSTRACT

Relation-aware graph structure embedding (RaGSE) is promising for predicting multi-relational drug–drug interactions (DDIs). Most existing methods based on RaGSE commence by constructing a multi-relational DDI graph. They then learn the RaGSEs for drugs by aggregating their neighbor's features under different relations. Nevertheless, new drugs lack neighbors in the DDI graph. This limitation hinders the ability of these methods to effectively learn RaGSEs for new drugs, resulting in suboptimal performance when evaluating DDIs that involve new drugs. To alleviate this issue, we propose a novel DDI prediction method based on relation-aware graph structure embedding with co-contrastive learning, RaGSECo. The proposed RaGSECo constructs two heterogeneous graphs: a multi-relational DDI graph and a multi-attribute drug–drug similarity (DDS) graph. The two graphs are used respectively for learning and propagating the RaGSEs of drugs, aiming to ensure all drugs, including new ones, can possess effective RaGSEs. Additionally, we present a novel co-contrastive learning module to learn feature representations for drug pairs (DPs). This module learns DP representations from two distinct views (interaction and similarity views) and encourages these views to supervise each other collaboratively to obtain more discriminative DP representations. We evaluate the effectiveness of our RaGSECo on three different tasks using two real datasets. Experimental outcomes unequivocally indicate that our RaGSECo surpasses several state-of-the-art methods in performance.

1. Introduction

Human diseases are often the result of complex biological processes, and single-drug treatments are often insufficient to cure them entirely [1,2]. Consequently, combination drug therapy has become attractive as it can reduce drug resistance and improve efficacy [3,4]. Nevertheless, the simultaneous use of multiple drugs can alter their properties and lead to adverse drug interactions, which can cause harm to patients [5]. Therefore, identifying potential DDIs is crucial for safe coadministration [6]. Traditional methods for detecting DDIs, such as biological or pharmacological assays, are labor-intensive, time-consuming, and expensive [7,8]. In contrast, computation-based methods are generally less expensive and can achieve higher accuracy than traditional methods [9].

Recently, many computational methods have been proposed for predicting DDI events. For instance, Yu et al. [10] utilized triplets (substructure, substructure, interaction type) to construct a substructure–substructure interaction (SSI) graph. From this, they learned drug embeddings for multi-relational DDI prediction. Wang et al. [11] incorporated various biochemical attributes and graph convolutional layers to derive drug representations for new DDI predictions. Pang et al. [12] employed the Simplified Molecular-Input Line-Entry System (SMILES) string and atomic graph of the drug to encode drug features for DDI prediction. Ren et al. [13] built a biomedical knowledge graph (BKG) and extracted local–global information from it to enhance DDI prediction performance. Zhao et al. [14] used biological knowledge to develop a heterogeneous information network (HIN), introducing a meta-path-based information fusion mechanism for learning discriminative drug features for multi-relational DDI prediction. Feng et al. [15] designed a social theory-enhanced graph representation learning framework to learn drug embeddings, aiming to predict DDIs more effectively.

Many researchers regard the drugs that possess known interactions as known drugs and refer to those without known interactions as new drugs [16]. Typically, the lack of interaction information on new drugs makes it difficult for models to predict corresponding interactions accurately [16]. Nevertheless, predicting new drug-involved DDIs is significant for the efficient development of new drugs [17]. Therefore, many researchers concern this task [18]. For instance, Deng et al. [16]

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utilized multiple biochemical attributes to calculate drug similarities as drug features and introduced a multimodal deep learning framework to predict DDIs for new drugs. Zhang et al. [19] integrated various biochemical attributes for drug feature extraction using sparse feature learning and then applied linear neighborhood regularization for DDI prediction. Lin et al. [20] merged three biochemical attributes to calculate drug similarities as features and developed multiple encoders to learn comprehensive DP representations. The aforementioned methods employ multiple biochemical attributes to learn drug features and have achieved leading performance when predicting new drug-involved DDIs. Nevertheless, to ensure that the training and test data have the same distribution, these methods did not use multi-relational interaction information between drugs, limiting the DDI prediction performance.

Recently, self-supervised learning, which aims to derive supervised signals from the data itself spontaneously, has emerged as a promising strategy for effective representation learning [21]. Among the various techniques under the umbrella of self-supervised learning, contrastive learning has attracted substantial attention [22]. Contrastive learning first extracts positive and negative samples from data and then maximizes the similarities between positive samples while minimizing the similarities between negative samples [23]. In this way, contrastive learning can learn discriminative representations even without abundant labels. Despite the broad application of contrastive learning in computer vision [24] and natural language processing [25], its potential in the context of DDI prediction tasks has rarely been explored. Implementing contrastive learning to facilitate DDI prediction is by no means trivial and requires careful consideration of the task-specific characteristics and nuances of contrastive learning.

Based on the above discussion, the primary motivation for our work lies in enabling all drugs to distill effective RaGSEs, thereby facilitating DP representation learning and improving DDI prediction. To achieve this goal, we propose a relation-aware graph structure embedding with co-contrastive learning (RaGSECo) framework, an end-to-end learning model. Implementing our RaGSECo approach includes two main stages: RaGSE and co-contrastive learning. The purpose of designing the RaGSE method is to learn effective RaGSEs for all drugs. In the RaGSE method, we construct a multi-relational DDI graph using known DDIs (training set). Within the DDI graph, known drugs aggregate their neighbor's features under different relations to obtain effective RaGSEs. However, new drugs, lacking known interactions, are devoid of neighbors in the DDI graph, rendering them incapable of acquiring effective RaGSEs. To address this challenge, we draw inspiration from the premise that drugs with similar properties may interact with the same set of drugs [16,20,26]. Consequently, we construct a multi-attribute DDS graph. Within this DDS graph, the embeddings for new drugs are improved by aggregating the RaGSEs of their neighboring drugs. This strategy ensures that all drugs, whether new or known, are equipped with effective RaGSEs. Furthermore, we incorporate multi-view representation learning and cross-view contrastive learning to present a novel co-contrastive learning mechanism to learn DP representations. Unlike previous contrastive learning, which contrasts the original and the corrupted representations, we design two distinct views for DP representation learning: interaction and similarity views. Specifically, we utilize the RaGSEs of drugs to generate the DP representations under the interaction view. Meanwhile, we leverage drug similarities to generate DP representations under the similarity view. The interaction view primarily utilizes known interaction relationships between drugs. The similarity view can discover implicit connections between drugs and facilitate inferring the potential therapeutic effects, toxicity reactions, or drug interactions of new drugs. Therefore, the two views are complementary. Moreover, considering underlying correlations between DPs, we design a novel positive selection strategy to enhance cross-view contrastive learning. With the training, these two views supervise each other collaboratively and learn more discriminative DP representations.

The main contributions of our work can be summarized as follows:

- We innovatively propose the strategy of propagating the RaGSEs of known drugs to new drugs with similar features. This advancement notably improves the DDI prediction performance of relation-aware-based methods in new drug scenarios.

- To our best knowledge, this is the first attempt to perform DP-level cross-view contrastive learning. More discriminative DP representations can be learned by co-contrastive learning based on a cross-view manner.

- The proposed RaGSECo ingeniously contrasts and incorporates two views of the drug information graph (interaction and similarity views), enabling this model to capture both the known and the potential interactions.

2. Related work

We review previous studies relevant to this work in three areas: relational graph neural networks, contrastive learning, and drug feature extraction.

2.1. Relational graph neural networks

The relational graph neural networks-based methods pay attention to the graph's topology, typically learning entity embeddings by aggregating information from neighboring entities under different relations [27]. The practicability of these methods on multi-relational DDI predictions makes these methods attract considerable attention. Most existing methods use graph neural networks (GNNs), including relational graph convolutional networks (R-GCNs) [28], graph attention networks (GATs) [29], and GraphSage [30], to learn drug representations. For example, Hong et al. [31] constructed different attention pathways between drugs based on different semantic relationships, using a GAT to learn drug embeddings by aggregating neighbor features from these diverse pathways. Yu et al. [27] built a multi-relational DDI graph and employed the GraphSage aggregator [30] to learn drug embeddings from the DDI graph. Wang et al. [32] utilized a GCN [33] to learn representations for atomic graphs of drugs and employed a GAT to obtain drug representation from the multi-relational DDI graph. Feng et al. [34] built a multi-relational DDI graph and applied GAT to learn the topological features of the DDI graph, thereby efficiently predicting DDI types for new drugs. The aforementioned methods take advantage of interaction information and perform well in the multi-relational DDI prediction task. Nevertheless, the absence of interaction information on new drugs limits the performance of predicting DDIs involving new drugs.

2.2. Contrastive learning

Contrastive learning contrasts positive pairs against negative pairs have achieved considerable success across various domains [35]. In this section, we mainly review the contrastive learning-based DDI prediction works. Zhao et al. [23] constructed an original and a corrupted DDI graph. They then minimized the mutual information between outputs of the original and corrupted networks and maximized the mutual information between outputs from either the original or the corrupted network alone. Zhang et al. [36] developed three distinct perspectives: molecular, structural, and semantic. They then enhanced the alignment between these perspectives through contrastive learning techniques. Gao et al. [37] designed the drug-disease association view and drug similarity view. Their approach focused on maximizing the mutual information across these two views. Zhuang et al. [38] leveraged DDI data to identify positive and negative samples. Subsequently, they fine-tuned their model parameters by optimizing the mutual information between these identified samples. Although many works have been proposed to improve drug embedding by contrastive learning, little effort has been made toward DP-level representation learning. Nevertheless, learning high-level DP representations is more practical for the DDI prediction task.

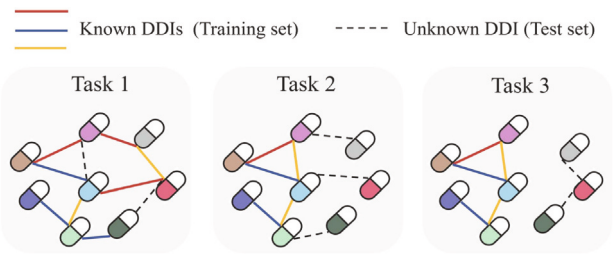


Fig. 1. Examples of construction strategies for three different tasks. The nodes represent drugs, and solid edges indicate known DDI interactions (training set). The edges with different colors signify various interaction event types. These dotted edges represent the prediction task (test set).

2.3. Drug feature extraction

Drug feature extraction is crucial for model training [20]. Zhu et al. [39] took into account eight biochemical attributes (molecular substructure, target, enzyme, pathway, side effect, phenotype, gene, and disease) to extract drug features. Deng et al. [16] and Lin et al. [20] utilized targets, enzymes, and chemical substructures to extract drug features. Typically, each biochemical attribute is associated with a set of descriptors. A drug can be denoted as a binary feature vector where the element (1 or 0) indicates the presence or absence of the corresponding descriptor [20]. Consequently, these sparse binary feature vectors typically have high dimensionalities. In general, high dimensional inputs can be resource-intensive and induce the curse of dimensionality, leading to poor performance in some cases [20]. There exists an assumption that similar drugs may interact with the same drugs [40]. Therefore, Deng et al. [16] and Lin et al. [20] opted to calculate the Jaccard similarities under different biochemical attributes to define drug features. The Jaccard similarity is calculated as follows:

$$\text{Jaccard}(U,V)=\frac{|U\cap V|}{|U\cup V|} \tag{1}$$

where U and V are the descriptors of two drugs under a specific biochemical attribute. Herein, $|U\cap V|$ is the cardinality of the intersection between U and V, and $|U\cup V|$ is the union.

3. Methodology

3.1. RaGSECo framework

The RaGSECo is proposed for multi-relational DDI prediction, which can be regarded as a multi-class classification task. Fig. 1 vividly exhibits three different multi-relational DDI prediction tasks. As the figure illustrates, in Task 1, all drugs involved in the test set appear in the training set. In Task 2, half of the drugs involved in the test set appear in the training set. In Task 3, the training and test sets possess different drugs. Therefore, from Task 1 to Task 3, the prediction difficulties increase in turn.

Fig. 2 presents the framework of our RaGSECo, using Task 3 as an example. RaGSECo constructs two heterogeneous graphs: a multi-relational DDI graph and a multi-attribute DDS graph. In the multi-attribute DDS graph, nodes denote drugs, and edges represent similarities between drugs. The edge colors indicate similarities under varying attributes (targets, enzymes, and chemical substructures). Consequently, in the DDS graph, each pair of drugs can have up to three types of edges. In contrast, in the DDI graph, each pair of drugs has only up to one type of edge. The method of extracting the initial features for drugs is the same as in [20], described in Section 2.3.

3.2. RaGSE

The RaGSE method is proposed to learn effective RaGSEs for all drugs. RaGSE consists of two parts: RaGSE learning (RaGSEL) and RaGSE propagation (RaGSEP). RaGSEL learns the RaGSEs for known drugs, and RaGSEP propagates the RaGSEs of known drugs to new drugs with similar features.

3.2.1. RaGSEL

We construct a multi-relational DDI graph $\mathcal{G}=(D,\mathcal{E},\mathcal{R})$, where nodes represent drugs, and edges denote labeled interactions. Herein, D represents the set of all drugs (including known and new drugs), and $|D|=N$ denotes the number of drugs. Let $\mathbf{X}=\{\mathbf{x}_i\}_{i=1}^{N}\in\mathbb{R}^{N\times d}$ be the initial feature matrix of drugs. $\mathcal{R}$ denotes the set of interaction event types, and $|\mathcal{R}|=R$ represents the number of interaction event types. $\mathcal{E}=\{\mathcal{E}_r\}_{r=1}^{R}$ denotes the set of known interactions, and $\mathcal{E}_r$ represents the set of interactions under interaction event type r. Let $\mathcal{A}=\{\mathbf{A}_r\}_{r=1}^{R}\in\mathbb{R}^{N\times N\times R}$ be the multi-relational adjacency tensor, where $\mathbf{A}_r\in\mathbb{R}^{N\times N}$ represents the adjacency matrix under interaction relation r. Let $A_{r(i,j)}$ $(i,j=1,\ldots,N)$ be the element of $\mathbf{A}_r$, $A_{r(i,j)}=1$ if $(i,j)\in\mathcal{E}_r$ and $A_{r(i,j)}=0$ if $(i,j)\notin\mathcal{E}_r$.

Subsequently, we employ the R-GCN layer [28] to learn the RaGSEs of drugs from the multi-relational DDI graph. The forward propagation function is defined as follows:

$$\mathbf{h}_i=\sigma\left(\sum_{r\in\mathcal{R}}\sum_{j\in\mathcal{N}_i^r}\frac{\hat{A}_{r(i,j)}}{R_i}\mathbf{x}_j\mathbf{W}_r+\mathbf{x}_i\mathbf{W}_o\right) \tag{2}$$

where $\mathbf{h}_i\in\mathbb{R}^{1\times d'}$ represents the RaGSEs for drug i, and $\mathbf{x}_i\in\mathbb{R}^{1\times d}$ are the initial feature vector of drug i, $i\in D$. Herein, $\mathbf{W}_r\in\mathbb{R}^{d\times d'}$ represents the relation-specific weight matrix and the adoption of a set of $\mathbf{W}_r$ $(r=1,\ldots,R)$ supports multiple edge types. On the contrary, $\mathbf{W}_0\in\mathbb{R}^{d\times d'}$ is a single weight matrix regardless of relations. Here, $\mathcal{N}_i^r$ represents the set of drugs connected to drug i under relation r. $\sigma(\cdot)$ is an element-wise activation function: $\text{ReLU}(\cdot)=\max(0,\cdot)$. To normalize the incoming messages of each drug, we use $\{R_i\}_{i=1}^{N}$ to define a set of normalized constants. Here, R_i is a scalar and equals the number of interaction types in which drug i is involved. $\hat{A}_{r(i,j)}$ is the aggregation weight assigned to the neighboring drug j. Specifically, $\hat{A}_{r(i,j)}$ denotes the element in $\hat{\mathbf{A}}_r$. The calculation of $\hat{\mathbf{A}}_r$ is based on a classic graph-based normalization method [33].

$$\hat{\mathbf{A}}_r=\mathbf{D}_r^{-1/2}\mathbf{A}_r\mathbf{D}_r^{-1/2} \tag{3}$$

where $\mathbf{D}_r=\text{diag}\left(\mathbf{A}_r\mathbf{1}_N\right)$ is the degree matrix of $\mathbf{A}_r$.

When drug i is known, $\mathbf{h}_i$ can represent relation-aware graph structure information for drug i. However, when drug i is new, drug i has no neighbors in the DDI graph. Thus, $\mathbf{h}_i$ would be influenced only by drug i itself and cannot represent effective RaGSEs for drug i. Directly using the output of Eq. (2) as the drug embeddings could result in poor performance on Tasks 2 and 3.

3.2.2. RaGSEP

Based on the assumption that similar drugs may interact with the same drugs [16,20], we propose a strategy that propagates the RaGSEs of known drugs to new drugs with similar features. This strategy can effectively improve new drugs' RaGSEs and enhance the generalization and robustness of the model, especially in Tasks 2 and 3. To facilitate the propagation of embeddings, we construct a multi-attribute DDS graph $\mathcal{G}_s=(\mathbf{H},\mathcal{A})$, where nodes represent drugs, and edges denote Jaccard similarities between drugs under various attributes. Herein, we still employ these three biological attributes (i.e., chemical substructure, enzyme, and target) to calculate drug similarities. In the DDS graph, $\mathbf{H}=\{\mathbf{h}_i\}_{i=1}^{N}$ represents the feature matrix of nodes, which is the output of Eq. (2). $\mathcal{A}=\{\mathbf{A}_s,\mathbf{A}_e,\mathbf{A}_t\}\in\mathbb{R}^{N\times N\times 3}$ represents the adjacency tensor, where $\mathbf{A}_s$, $\mathbf{A}_e$, and $\mathbf{A}_t$ denotes the substructure-based, enzyme-based, and target-based adjacency matrix, respectively.

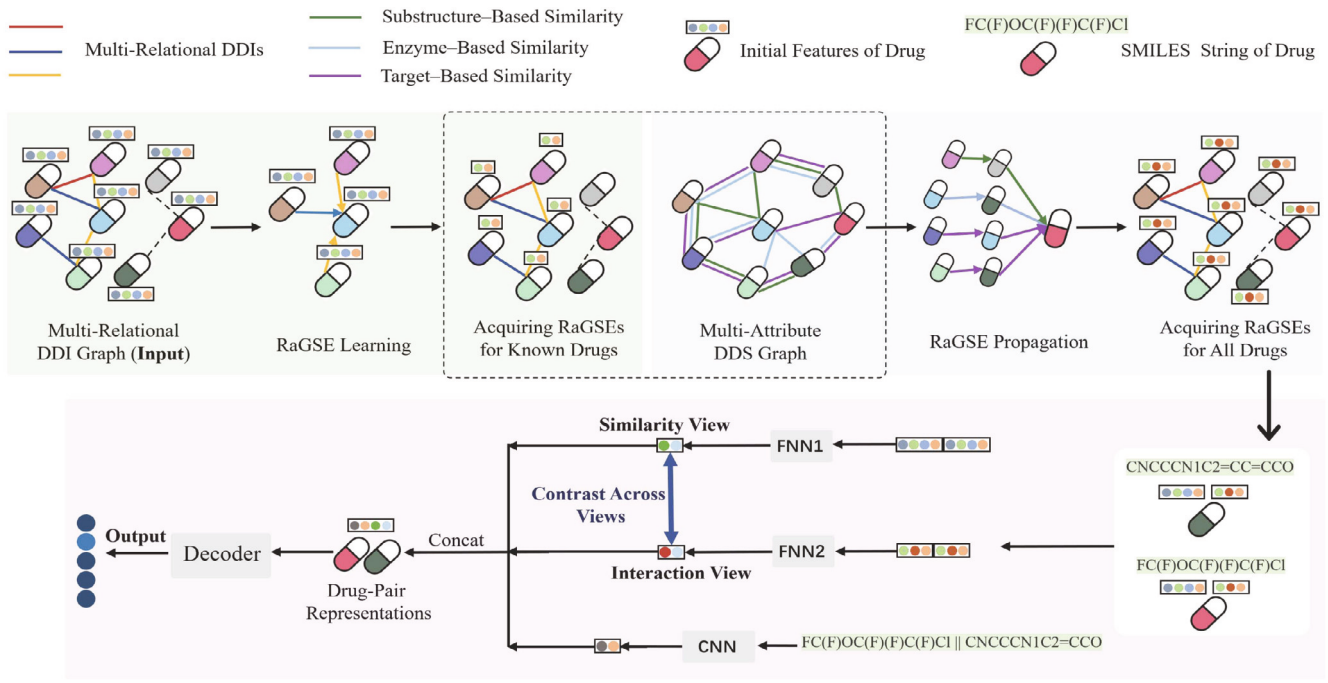


Fig. 2. Flowchart of the proposed RaGSECo, involving two main parts: RaGSE (**Upper**) and co-contrastive learning (**Bottom**). In the RaGSE method, we first learn the RaGSEs for known drugs by propagating their neighbor's features under different relations within the DDI graph. Subsequently, we propagate the learned RaGSEs to the connected drugs within the DDS graph to ensure that all drugs possess effective RaGSEs. In co-contrastive learning, the initial features of two drugs are concatenated and fed into FNN1, resulting in the DP representations under the similarity view. Likewise, the RaGSEs of two drugs are concatenated and fed into FNN2, generating the DP representations under the interaction view. Meanwhile, a Convolutional Neural Network (CNN) is employed to obtain the SMILES-based DP representations. Additionally, a cross-view contrastive mechanism facilitates collaborative supervision between similarity and interaction views, thereby promoting the learning of more discriminative DP representations. Finally, the three DP representations are concatenated and fed into a Decoder to calculate the classification probability distributions of the DP.

To normalize the incoming messages, these three adjacency matrices also need to be normalized. The normalization method is referred to as Eq. (3). Let $\hat{\mathbf{A}}_s$, $\hat{\mathbf{A}}_e$, and $\hat{\mathbf{A}}_t$ be the normalized adjacency matrices. We use these normalized adjacency matrices to propagate the learned RaGSEs to the connected drugs within the DDS graph. The propagation procedure can be expressed as follows:

$$\mathbf{H}_s = \sigma\left(\hat{\mathbf{A}}_s^n \mathbf{H}\mathbf{W}_s\right) \tag{4}$$

$$\mathbf{H}_e = \sigma\left(\hat{\mathbf{A}}_e^n \mathbf{H}\mathbf{W}_e\right) \tag{5}$$

$$\mathbf{H}_t = \sigma\left(\hat{\mathbf{A}}_t^n \mathbf{H}\mathbf{W}_t\right) \tag{6}$$

$$\mathbf{H}_{embed} = \mathbf{H}_s + \mathbf{H}_e + \mathbf{H}_t \tag{7}$$

where $\hat{\mathbf{A}}_s^n\mathbf{H}$, $\hat{\mathbf{A}}_e^n\mathbf{H}$, and $\hat{\mathbf{A}}_t^n\mathbf{H}$ represent graph convolutions on the DDS graph. n is a hyperparameter, and $\hat{\mathbf{A}}_s^n$ is $\hat{\mathbf{A}}_s$ to the power of n. The same definition is extended to $\hat{\mathbf{A}}_e^n$ and $\hat{\mathbf{A}}_t^n$. This strategy allows RaGSECo to capture higher-order neighbor information flexibly. $\mathbf{W}_s$, $\mathbf{W}_e$, $\mathbf{W}_t \in \mathbb{R}^{d' \times d'}$ are the attribute-specific trainable weight matrices. $\mathbf{H}_s$, $\mathbf{H}_e$, and $\mathbf{H}_t$ are the propagation results along with different pathways. $\mathbf{H}_{embed} = \{\mathbf{h}_{embed,i}\}_{i=1}^{N} \in \mathbb{R}^{N \times d'}$ is the sum of the three propagation results and represents the embedding matrix of drugs. Particularly, $\mathbf{h}_{embed,i}$ denotes effective RaGSEs for drug i, whether drug i is known or new.

3.3. Co-contrastive learning

The co-contrastive learning module is designed to learn DP representations. This module consists of two primary components: multi-view representation learning and collaborative contrastive optimization. The first component learns DP representations from multiple views. The second component facilitates a cooperative process in which these views optimize and supervise each other.

3.3.1. Multi-view representation learning

To increase data diversity, facilitate gradient propagation, and generate discriminative DP representations, we employ multiple drug features, i.e., initial features, RaGSEs, and SMILES string, to learn DP representations.

SMILES string is a line notation that uses a predefined set of rules to describe the structure of compounds, and each drug has its own SMILES string [41]. The characters of a SMILES string represent chemical atoms or chemical bonds, and the lengths of the SMILES strings are unconstrained [42]. In this work, we convert each SMILES string to a $p \times q$ dimensional feature matrix for simplicity, where p denotes the number of characters and q represents the unified length of the SMILES string [43]. Accordingly, the columns of the SMILES-based feature matrix are one-hot vectors. Since the initial lengths of the SMILES strings are inconsistent, we will cut off the extra part if the actual length of the SMILES string is greater than q, and we will pad zeros if the actual length is less than q. The p and q are set as 64 and 100 according to Huang et al. [43].

Considering computational burdens, we adopt a batch-wise scheme to train the RaGSECo. Let K be the number of DPs in a batch. Given DP k ($k = 1, \ldots K$) that involves two drugs i and j ($i, j = 1, \ldots N; i \neq j$). $\mathbf{x}_i$ and $\mathbf{x}_j$ denote initial feature vectors of the two drugs. $\mathbf{h}_{embed,i}$ and $\mathbf{h}_{embed,j}$ represent the RaGSEs of the two drugs. Let $\mathbf{S}_i$, $\mathbf{S}_j \in \mathbb{R}^{p \times q}$ be the SMILES-based feature matrices of the two drugs. Subsequently, we employ two feedforward neural networks (FNNs) to encode the RaGSEs and initial features of DPs, respectively. In addition, a CNN encodes SMILES strings of DPs. The employed CNN is a multi-layer 1-D CNN followed by a global max pooling layer, and with reference to [43]. Accordingly, three features of DP k are encoded as follows:

$$\mathbf{p}_k^{initi} = \mathrm{FNN1}(\mathbf{x}_i \parallel \mathbf{x}_j, \Theta_{\mathrm{FNN1}}) \tag{8}$$

$$\mathbf{p}_k^{embed} = \mathrm{FNN2}(\mathbf{h}_{embed,i} \parallel \mathbf{h}_{embed,j}, \Theta_{\mathrm{FNN2}}) \tag{9}$$

$$\mathbf{p}_k^{smile} = \mathrm{CNN}(\mathbf{S}_i \parallel \mathbf{S}_j, \Theta_{\mathrm{CNN}}) \tag{10}$$

where the symbols Θ_{FNN1}, Θ_{FNN2}, and Θ_{CNN} represent the trainable weights involved in FNN1, FNN2, and CNN, respectively.

The initial feature vectors of drugs are denoted by Jaccard similarity scores between drugs. Thus, $\mathbf{p}_k^{initi} \in \mathbb{R}^{1\times d^{FNN}}$ is considered as the feature representations for DP i under the similarity view. The drug RaGSEs primarily concentrate on interaction information between drugs. As such, $\mathbf{p}_k^{embed} \in \mathbb{R}^{1\times d^{FNN}}$ is perceived as the DP representations under the interaction view.

3.3.2. Collaboratively contrastive optimization

The interaction view mainly focuses on known interaction relationships. In contrast, the similarity view can employ drug correlations to infer potential therapeutic effects, adverse reactions, or interactions involving new drugs. Therefore, these two views are complementary and mutually reinforcing. Herein, we define positive and negative samples for meeting contrastive learning. In this paper, given the feature representations of a DP under the interaction view, we can simply define its feature representations under the similarity view as the positive sample. Nevertheless, our analysis of the employed datasets suggests that there may be underlying correlations between DPs. Consequently, we propose a novel positive selection strategy.

This paper utilizes two real-world, multi-relational DDI datasets: Dataset 1 and Dataset 2. In Dataset 1, the known DDIs are categorized into 65 types of DDI events, while the number of interaction event types in which each drug is involved ranges from 1 to 20. The median of the type counts is 10, significantly lower than 65. Similar findings are also observed in Dataset 2. These findings suggest that a particular drug is likely to associate with specific interaction event types strongly. Thus, the interaction event types a drug participates in can be considered significant characteristics of this drug. Herein, we refer to these significant characteristics as interaction characteristics of drugs. Let $\mathbf{t}_i \in \mathbb{R}^{1\times R}$ represent the interaction characteristics of drug i, where R is the number of interaction event types. The vector $\mathbf{t}_i$ is binary, and its nonzero elements represent the interaction event types the drug i participates in. Given a DP k that involves two drugs i and j, the interaction characteristics of DP k are defined as $\mathbf{c}_k = \mathbf{t}_i + \mathbf{t}_j$. Based on this, we propose a new positive selection strategy: if two DPs exhibit similar interaction characteristics, they are designated as positive samples. One advantage of such a strategy is that the selected positive samples may reflect the underlying interaction tendency of the target DP. For two DPs k and q, we calculate the cosine similarity of their interaction characteristics as follows:

$$S_{k,q} = \frac{\mathbf{c}_k \cdot \mathbf{c}_q}{\|\mathbf{c}_k\|\|\mathbf{c}_q\|}. \tag{11}$$

Given the threshold values t_{pos} and t_{neg}, where $t_{pos} > t_{neg}$. DP q is deemed a positive sample of DP k if $S_{k,q} \geq t_{pos}$, and conversely, is considered a negative sample if $S_{k,q} \leq t_{neg}$. Let $\mathcal{P}$ represent the set of positive pairs of one batch, and $\mathcal{N}$ be the set of negative pairs. We then model a co-contrastive learning task using the standard binary cross-entropy loss. With this, we present the co-contrastive learning loss function as follows:

$$\ell_{co1} = -\frac{1}{|\mathcal{P}| + |\mathcal{N}|}\left(\sum_{(k,q)\in\mathcal{P}} \log \Psi\left(\mathbf{p}_k^{embed}, \mathbf{p}_q^{initi}\right) + \sum_{(k,q)\in\mathcal{N}} \log\left(1 - \Psi\left(\mathbf{p}_k^{embed}, \mathbf{p}_q^{initi}\right)\right)\right) \tag{12}$$

where the contrastive discriminator $\Psi(\cdot,\cdot)$ is instantiated as $\sigma(\mathbf{p}_k^{embed}\mathbf{W}\mathbf{p}_q^{initi^T})$. $\mathbf{W}$ is a learnable weight matrix. The activation function σ is the sigmoid function, which produces a score representing the probability of being a positive sample. The loss function ℓ_{co1} pursues that maximizes the mutual information of positive pairs and minimizes the mutual information of negative pairs. To stabilize the co-contrastive learning process and facilitate two views supervising each other, we present another co-contrastive learning loss function:

$$\ell_{co2} = -\frac{1}{|\mathcal{P}| + |\mathcal{N}|}\left(\sum_{(k,q)\in\mathcal{P}} \log \Psi\left(\mathbf{p}_k^{initi}, \mathbf{p}_q^{embed}\right) + \sum_{(k,q)\in\mathcal{N}} \log\left(1 - \Psi\left(\mathbf{p}_k^{initi}, \mathbf{p}_q^{embed}\right)\right)\right). \tag{13}$$

To fully utilize all significant information and learn more comprehensive DP representations, we calculate the feature representations for DP k as follows:

$$\mathbf{p}_k^{add} = \mathbf{p}_k^{initi} + \mathbf{p}_k^{embed} + \mathbf{p}_k^{smile} \tag{14}$$

$$\mathbf{p}_k = \parallel\left(\mathbf{p}_k^{initi}, \mathbf{p}_k^{embed}, \mathbf{p}_k^{smile}, \mathbf{p}_k^{add}\right) \tag{15}$$

where the symbol $\parallel$ denotes the concatenation.

3.4. Loss function

We construct a Decoder by two fully connected layers that map $\mathbf{p}_k$ into the probability distribution space $\mathbf{y}_k \in \mathbb{R}^{1\times R}$:

$$\mathbf{y}_k = \mathrm{Decoder}\left(\mathbf{p}_k; \Theta_{\mathrm{Decoder}}\right) \tag{16}$$

where $\Theta_{\mathrm{Decoder}}$ is the trainable parameters. The first fully connected layer is followed by a ReLu activation function and a dropout layer. The second fully connected layer is followed by a softmax function.

DDI events prediction is a multi-classification task. Thus, we choose the cross-entropy loss function to calculate classification loss:

$$\mathcal{L}_{ce} = -\sum_{k=1}^{K}\sum_{r=1}^{R} \hat{y}_{k,r} \log\left(y_{k,r}\right) \tag{17}$$

where $\hat{y}_{k,r}$ represents the rth element of $\hat{\mathbf{y}}_k$, which corresponds to the ground-truth vector (one-hot encoding) of the DP k. K denotes the number of DPs in a single batch for training. On the other hand, $y_{k,r}$ denotes the predicted probability score of the DP k under the class r, corresponding to the rth element of $\mathbf{y}_k$. To emphasize the importance of the classification loss, we introduce a hyperparameter λ to scale the classification loss $\mathcal{L}_{ce}$. Accordingly, the total loss function of the model is formulated as follows:

$$\mathcal{L} = \lambda\mathcal{L}_{ce} + \mathcal{L}_{co1} + \mathcal{L}_{co2} \tag{18}$$

where $\mathcal{L}$ is the loss of one batch. Furthermore, we employ Mixup [44], a data augmentation algorithm, to increase the quantity of the original data and improve the generalization ability and robustness of the model. Mixup is described in detail in the previous study [44].

RaGSECo is a semi-supervised DDI prediction approach. Both known and new drugs are incorporated into each task's training and testing phases. We present Pseudocode in Algorithm 1 for a clearer illustration of optimizing the stated objective. We begin by extracting drug features via Jaccard similarities (Line 1). We then construct a multi-relational DDI graph using known DDIs (Line 2). During each batch of each training iteration, our initial step is to learn embeddings $\mathbf{H}_{embed}$ for all drugs (Lines 7–9). Then, a batch of training data is obtained from $\mathcal{T}_r$, and feature representations $\mathbf{P}_k$ for the corresponding DPs are learned (Lines 11, 12, and 15). In addition, we construct sets of positive and negative samples and calculate co-contrastive learning losses $\mathcal{L}_{co1}$, $\mathcal{L}_{co2}$ (Line 13 and 14). The classification loss, denoted as $\mathcal{L}_{ce}$, is subsequently computed (Lines 17 and 18). In the following step, we merge $\mathcal{L}_{co1}$, $\mathcal{L}_{co2}$ and $\mathcal{L}_{ce}$ to generate the final loss $\mathcal{L}$ (Line 19), then optimize $\mathcal{L}$ to update $\mathbf{H}$, $\mathbf{H}_{embed}$ and $\mathbf{P}_k$ (Line 20). Once all training iterations are complete, we use the trained RaGSECo to test the interaction event types of DPs in $\mathcal{T}_e$.

Algorithm 1: Pseudocode of the proposed RaGSECo.

Input: N_{epochs}: maximum epochs; $N_{batches}$: total batches; λ: hyperparameter; $\mathcal{T}_r$: the set of training DDIs; $\hat{\mathbf{Y}}$: ground truth of training DDIs; $\mathcal{T}_e$: the set of test DDIs.
Output: $\mathbf{Y_{te}}$: predicted types of test DDIs.

$\mathbf{X} \leftarrow$ Obtain drugs' initial features via (1).
$\mathcal{G} \leftarrow$ Construct a DDI graph using known DDIs and $\mathbf{X}$.
//∗ Training Procedure ∗//
for $epoch \in [1, N_{epochs}]$ **do**
  for $i \in [1, N_{batches}]$ **do**
    //∗ Acquiring Drug Embeddings∗//
    Use $\mathcal{G}$ learn RaGSEs of known drugs, $\mathbf{H}$, via Eq. (2).
    Use Jaccard similarities of drugs and $\mathbf{H}$ to build a DDS graph $\mathcal{G}_s$.
    Use $\mathcal{G}_s$ obtain RaGSEs for all drugs, $\mathbf{H}_{embed}$, via Eq. (4)-(7).
    //∗ Acquiring Drug-Pair Representations∗//
    Obtain ith batch training DDIs from $\mathcal{T}_r$, and let the DP k belong to the ith batch training DDIs.
    Acquire three representations of DP k: $\mathbf{p}_k^{initial}$, $\mathbf{p}_k^{embed}$, $\mathbf{S}_k^{smile}$, via Eq. (8)-(10).
    Obtain positive and negative sample set via Eq. (11).
    $\mathcal{L}_{co1}$, $\mathcal{L}_{co2} \leftarrow$ obtain co-contrastive learning losses via Eq. (12) and Eq. (13).
    Calculate final representations of DP k, $\mathbf{p}_k$, via Eq. (14) and Eq. (15).
    //∗ DDI Prediction∗//
    Use $\mathbf{p}_k$ to calculate the probability distribution $\mathbf{y}_k$ via Eq. (16).
    $\mathcal{L}_{ce} \leftarrow$ use ground truth $\hat{\mathbf{y}}_\mathbf{k}$ and predicted result $\mathbf{y}_k$ calculate prediction loss via Eq. (17).
    Obtain the final loss $\mathcal{L} \leftarrow \lambda\mathcal{L}_{ce} + \mathcal{L}_{co1} + \mathcal{L}_{co2}$.
    Minimize $\mathcal{L}$ and update $\mathbf{H}$, $\mathbf{H}_{embed}$, and $\mathbf{p}_k$.
  end
end
//∗ Testing Procedure ∗//
Predict types of the DDIs in $\mathcal{T}_e$ using the trained RaGSECo model.

4. Results and discussion

4.1. Datasets

This paper uses two real datasets to explore the effectiveness and competitiveness of our RaGSECo. The first data set (Dataset 1) was collected by Deng et al. from DrugBank[1] and published in [16]. Dataset 1 contains 572 drugs with 37 264 pairwise DDIs associated with 65 interaction types. Each drug is represented by four biological attributes: enzymes, targets, pathways, and chemical substructures.

We also construct a real dataset (Dataset 2) from DrugBank. We select 1000 drugs, each with three features: enzymes, targets, and chemical substructures. Within the 1000 drugs, we collect 206 029 pairwise DDIs associated with 99 types of interaction events. Dataset 2 can enhance the effectiveness of model training, but it also presents challenges due to the greater diversity of DDI event types. Essentially, Dataset 2 serves as an evaluation platform for assessing performance on a large-scale dataset. The detailed information on the two datasets is listed in Table 1.

[1] https://go.drugbank.com/releases/latest

Table 1
Summary of the datasets utilized in our experiments.

Dataset	Drug number	DDI number	DDI event types
Dataset 1	572	37 264	65
Dataset 2	1000	206 029	99

4.2. Baselines

We compare the proposed RaGSECo with six outstanding baselines:

- MM-GANN-DDI [45] is a multimodal graph-agnostic neural network that mines six drug modalities and employs GNN to capture topological information from the DDI graph and efficiently predicts DDI types for new drugs.
- MCFF-MTDDI [46] is a gated recurrent unit-based multi-channel feature fusion module that yields comprehensive representations of DPs for DDI prediction.
- MAEDDIE [47] is an attention-mechanism-based encoder–decoder model that constructs a multisource feature fusion network to learn drug embeddings for DDI prediction.
- MDF-SA-DDI [20] develops four encoders to generate DP features and adopts transformers to perform DP-level feature fusion for DDI prediction.
- RANEDDI [27] is a relation-aware graph structure embedding-based method that considers the multi-relational interaction information to obtain drug embedding.
- DDIMDL [16] calculates multiply drug–drug-similarities as drug features and developed a multi-modal deep learning framework to perform multi-relational DDI prediction.

4.3. Experimental settings

To demonstrate the effectiveness, we evaluate our RaGSECo by implementing Tasks 1, 2, and 3 on the two real datasets. In Task 1, we randomly split DDIs into five subsets via 5-fold cross-validation (5-CV), with four subsets as the training set and the remaining as the test set. For Tasks 2 and 3, we randomly split drugs instead of DDIs into five subsets via 5-CV, with four subsets used as the known drugs and the remaining as the new drugs. In Task 2, the DDIs among known drugs are defined as the training set, and the DDIs involving one known drug and one new drug are utilized as the test set. In Task 3, the training set is the same as Task 2, and the DDIs among new drugs are utilized as the test set.

The prediction task is the multi-class classification task. To evaluate the classification performance of our RaGSECo, we use six metrics: accuracy (ACC), the area under the precision–recall-curve (AUPR), the area under the ROC curve (AUC), Precision, Recall, and F1 score. The activation function and dropout layer are used between the fully connected layers. By default, we use the ReLU activation function and Radam optimizer [48]. The proposed RaGSECo is implemented with the deep learning library PyTorch. The Python and PyTorch versions are 3.8.10 and 1.9.0, respectively. All experiments are conducted on a Windows server with a GPU (NVIDIA GeForce RTX 4090).

4.4. Hyperparameter searching and setting

The proposed RaGSECo contains some hyperparameters that influence the prediction performance, including batch size (bs), learning

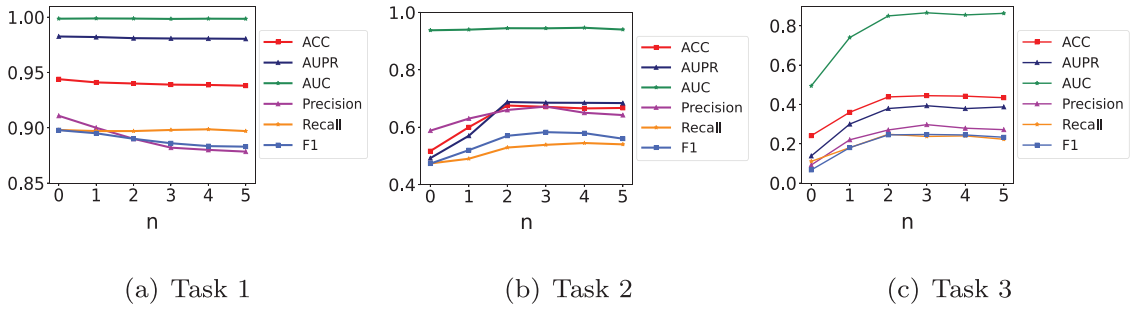


Fig. 3. Six metric scores versus the RaGSE propagation distance n on three different tasks of Dataset 1.

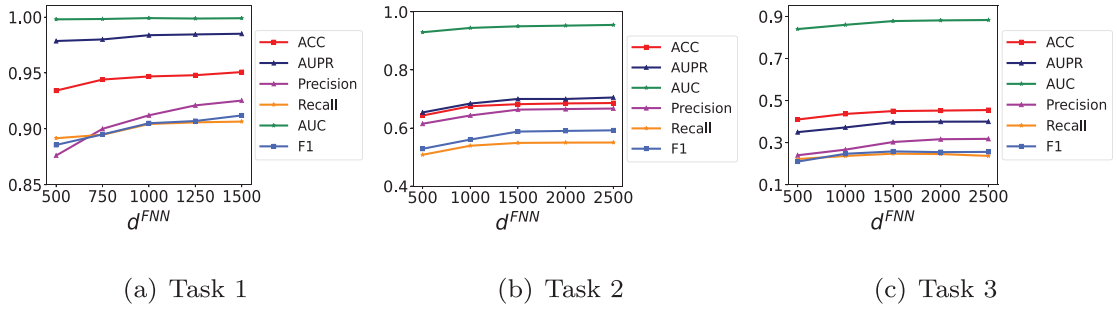


Fig. 4. Six metric scores versus the output dimension d^{FNN} of FNNs on three different tasks of Dataset 1.

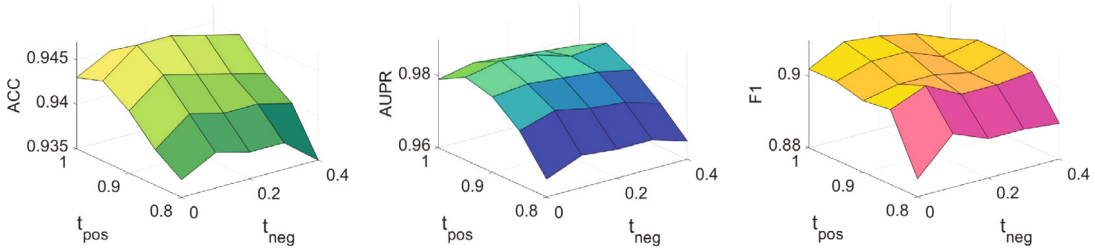


Fig. 5. ACC, AUPR, and F1 versus the threshold values t_{pos} and t_{neg} on Task 1 of Dataset 1.

rate (lr), dropout rate (dr), training epochs (te), the dimension of drug RaGSEs (d'), the power of adjacency matrices in the DDS graph (n), the output dimension of FNNs (d^{FNN}), threshold values (t_{pos} and t_{neg}), and the weight of cross-entropy loss (λ). To thoroughly investigate the impact of each parameter on the prediction results, a grid search strategy is employed where one parameter is varied while keeping the other parameters fixed. Table 5 summarizes the values of all hyperparameters on different datasets and tasks. Among these hyperparameters, n, d^{FNN}, t_{pos} and t_{neg} are specific to the proposed RaGSECo. Therefore, in the subsequent experiments, we will explore the impact of these hyperparameters on the experimental results and analyze the reasons for them.

4.4.1. Impact of ragse propagation distance n

The hyperparameter n represents the power of adjacency matrices in the DDS graph and determines the propagation distance of RaGSE. It plays a crucial role in generating effective RaGSEs for all drugs. To understand the influence of n on the prediction performance, we conduct experiments on three tasks of Dataset 1 and evaluate six metric scores across different n values. Referring to Fig. 3(a), in Task 1, the metric scores of RaGSECo lightly decrease as n increases. This could be because the effective relation-aware interaction information dilutes as n increases. Nevertheless, RaGSECo can consistently achieve good prediction performances. On the other hand, referring to Fig. 3(b) and (c), in Tasks 2 and 3, the prediction performance of RaGSECo is sensitive to the value of n. Specifically, when n is set to 0, we observe the lowest level of prediction performance. As n increases, there is a marked improvement in prediction accuracy. This trend is attributable to the nature of the test DDIs in Tasks 2 and 3, which involve new drugs. When n equals 0, these new drugs cannot aggregate useful information from similar drugs, leading to the suboptimal performance by RaGSECo. As n increases, the RaGSEs of known drugs are propagated to new drugs with similar properties, enhancing the embeddings of new drugs and significantly improving the generalization ability of RaGSECo. In addition, an optimal propagation distance helps in capturing proper contextual information. A propagation distance that is too long might cause an excessive dilution of information. Obviously, in Tasks 2 and 3, when n reaches 3, RaGSECo balances the extraction of local and global features, thereby archiving the highest metric scores. Based on the experimental results, we set n as 0 for Task 1. In Tasks 2 and 3, n is set as 3.

4.4.2. Impact of output dimension d^{FNN}

The output dimension d^{FNN} of FNNs is an important hyperparameter for our RaGSECo. Increasing d^{FNN} can enhance the generalization ability of RaGSECo to some extent. To investigate the impact of d^{FNN}

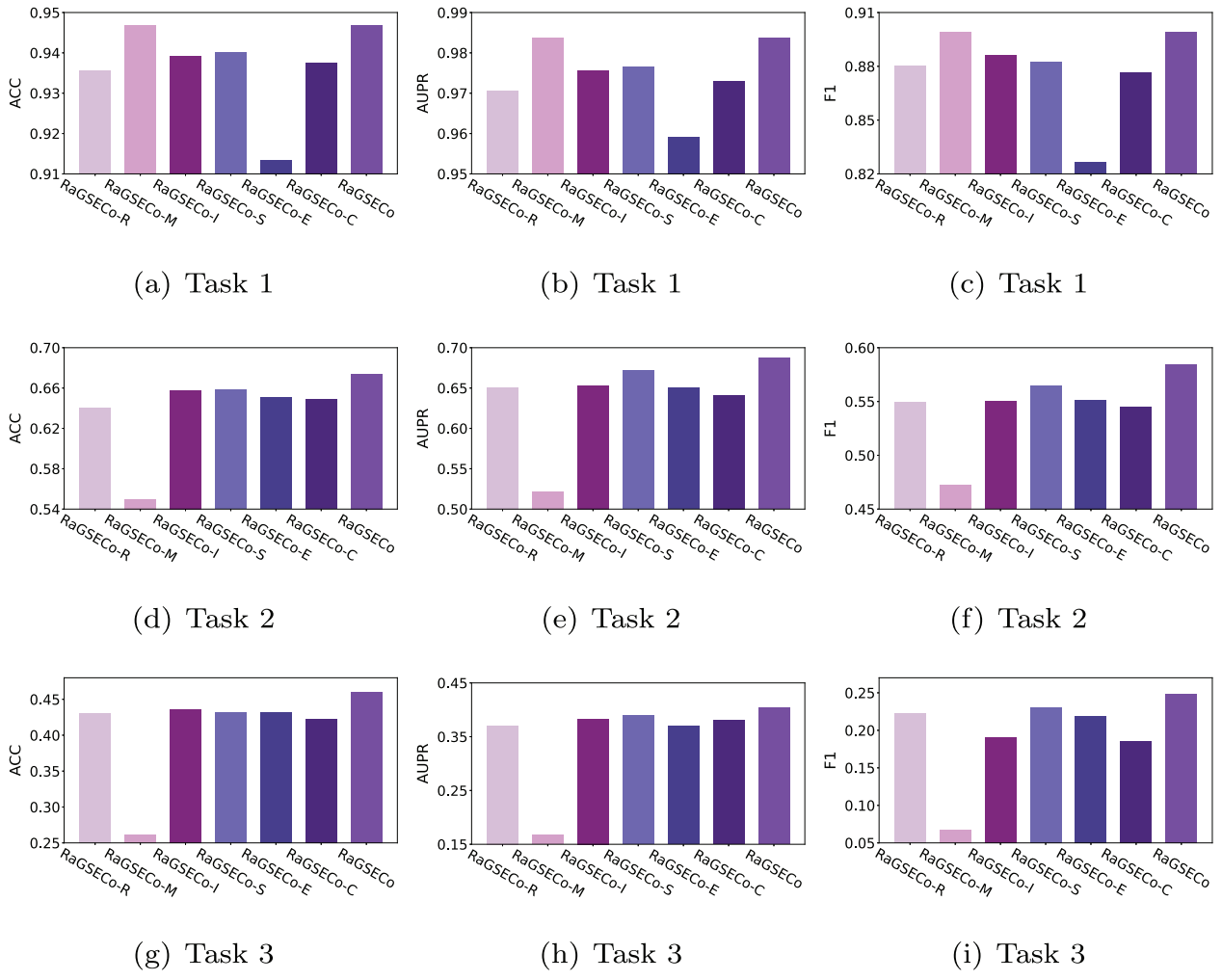


Fig. 6. Experimental results of RaGSECo and its six variants in terms of ACC, AUPR, and F1 on three tasks of Dataset 1. (a)–(c) Task 1. (d)–(f) Task 2. (g)–(i) Task 3.

on the prediction performance of RaGSECo, we conducted experiments on three tasks of Dataset 1. We examined the changes in six metric scores as d^{FNN} varies. From Fig. 4, it can be observed that metric scores gradually improve as d^{FNN} increase among all tasks. This discovery indicates the robustness of our model. To balance accuracy and efficiency, we set d^{FNN} to 1000 for Task 1. In Tasks 2 and 3, d^{FNN} is set as 1500.

4.4.3. Impact of threshold values t_{pos} and t_{neg}

The threshold values t_{pos} and t_{neg} are crucial for selecting proper positive and negative samples of DPs. In our experiments, we set t_{pos} to the range of $[0.8, 0.85, \dots, 1]$ and t_{neg} to the range of $[0, 0.1, \dots, 0.4]$. Fig. 5 presents the comparisons of three representative metric scores (ACC, AURR, and F1) for different values of t_{pos} and t_{neg}. It can be observed that the results are not significantly affected by variations in t_{pos} and t_{neg}. This indicates the stability of our co-contrastive learning strategy. Herein, the feature representations from the two views of each DP serve as positive samples for each other when t_{pos} equals 1. As observed, the model produces superior prediction results when t_{pos} is set to 0.95, as opposed to when it is set to 1. This observation indicates the effectiveness of our positive sample selection strategy. Finally, we set $t_{pos} = 0.95$ and $t_{neg} = 0.1$ for subsequent experiments.

4.5. Ablation study

In this subsection, we compare the proposed RaGSECo with six RaGSECo variants to assess the necessity and effectiveness of each component within RaGSECo. RaGSECo incorporates multiple components, including the construction of multi-relational DDI and multi-attribute DDS graphs, the integration of SMILES string, initial features, and RaGSEs of drugs, and the adoption of a co-contrastive learning mechanism. By comparing RaGSECo with its variants, we aim to understand the contributions of each component to the overall performance. The six variants are represented as follows:

- RaGSECo-R: A variant of RaGSECo that omits the multi-relational DDI graph for RaGSE learning. From a practical perspective, the initial features of drugs $\mathbf{X}$ are directly used as the node initial features of the multi-attribute DDS graph.
- RaGSECo-M: A variant of RaGSECo that omits the multi-attribute DDS graph for RaGSE propagation. In other words, the final drug embeddings are equal to the outputs of RaGSEL, $\mathbf{H}_{embed} = \mathbf{H}$.
- RaGSECo-I: A variant of RaGSECo that neglects the initial features of drugs during DP representation learning. Hence, the representations of DP k in Eq. (15) are redefined as: $\mathbf{p}_k = \| \left(\mathbf{p}_k^{smile}, \mathbf{p}_k^{embed}, \mathbf{p}_k^{smile} + \mathbf{p}_k^{embed}\right)$.
- RaGSECo-S: A variant of RaGSECo that does not use the SMILES strings of drugs during DP representation learning. Thus, the representations of DP k in Eq. (15) are redefined as: $\mathbf{p}_k = \| \left(\mathbf{p}_k^{initi}, \mathbf{p}_k^{embed}, \mathbf{p}_k^{initi} + \mathbf{p}_k^{embed}\right)$.
- RaGSECo-E: A RaGSECo variant that ignores drug RaGSEs during DP representation learning. Hence, the representations of DP k in Eq. (15) are redefined as: $\mathbf{p}_k = \| \left(\mathbf{p}_k^{initi}, \mathbf{p}_k^{smile}, \mathbf{p}_k^{initi} + \mathbf{p}_k^{smile}\right)$.
- RaGSECo-C: A RaGSECo variant that neglects the employment of co-contrastive learning. Hence, the representations of DP k in Eq. (15) are denoted as: $\mathbf{p}_k = \mathbf{p}_k^{embed}$. The total loss is represented as $\mathcal{L} = \mathcal{L}_{ce}$.

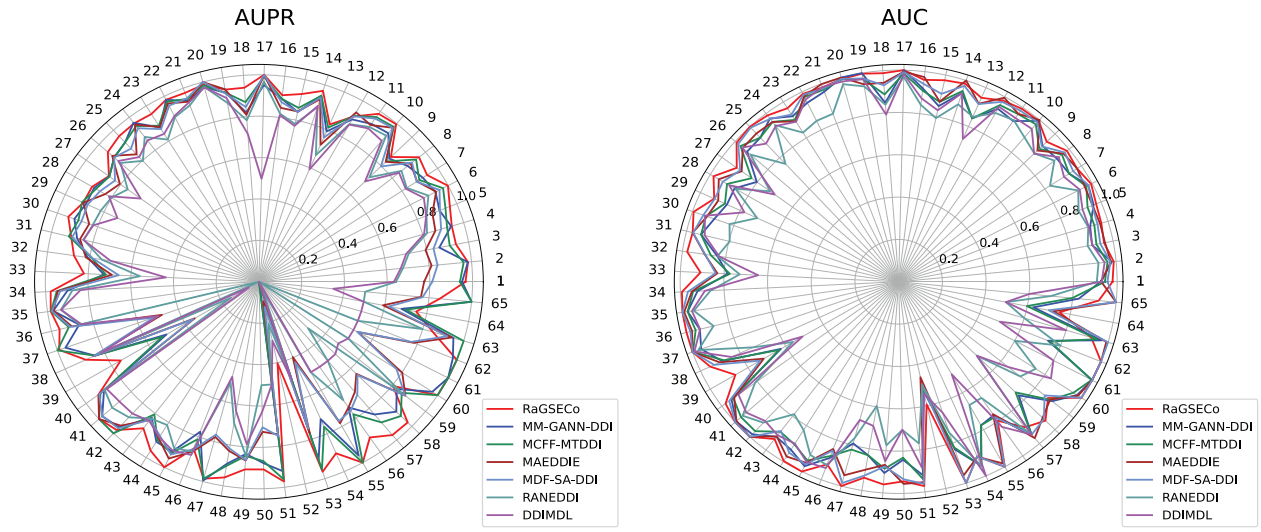


Fig. 7. Performance comparison for each DDI event type of Dataset 1.

Table 2
Prediction performances of different methods on Dataset 1.

	Method	ACC	AUPR	AUC	Precision	Recall	F1
Task 1	RaGSECo	**0.9461**	**0.9838**	**0.9991**	**0.9121**	**0.9043**	**0.9050**
	MM-GANN-DDI	0.9386	0.9786	0.9989	0.9088	0.8950	0.8980
	MCFF-MTDDI	0.9350	0.9757	0.9985	0.9100	0.8820	0.8918
	MAEDDIE	0.9426	0.9776	0.9989	0.9070	0.8781	0.8865
	MDF-SA-DDI	0.9301	0.9737	0.9989	0.9085	0.8760	0.8878
	RANEDDI	0.9228	0.9657	0.9980	0.8747	0.8701	0.8717
	DDIMDL	0.8852	0.9208	0.9976	0.8471	0.7182	0.7585
Task 2	RaGSECo	**0.6855**	**0.7115**	**0.9805**	0.6514	**0.5631**	**0.5860**
	MM-GANN-DDI	0.6705	0.6855	0.9532	0.6518	0.5156	0.5580
	MCFF-MTDDI	0.6650	0.6800	0.9500	**0.6561**	0.5139	0.5574
	MAEDDIE	0.6731	0.6893	0.9681	0.6552	0.5051	0.5565
	MDF-SA-DDI	0.6633	0.6776	0.9497	0.6547	0.5078	0.5584
	DDIMDL	0.6415	0.6558	0.9799	0.5607	0.4319	0.4460
Task 3	RaGSECo	**0.4591**	**0.4114**	**0.9548**	**0.3001**	**0.2513**	**0.2600**
	MM-GANN-DDI	0.4386	0.3786	0.8789	0.2774	0.2480	0.2505
	MCFF-MTDDI	0.4400	0.3870	0.8701	0.2823	0.2351	0.2437
	MAEDDIE	0.4510	0.3962	0.9271	0.2752	0.2251	0.2405
	MDF-SA-DDI	0.4338	0.3873	0.8630	0.2715	0.2226	0.2329
	DDIMDL	0.4075	0.3635	0.9512	0.2408	0.1452	0.1590

Table 3
Prediction performances of different methods on Dataset 2.

	Method	ACC	AUPR	AUC	Precision	Recall	F1
Task 1	RaGSECo	**0.9344**	**0.9805**	**0.9995**	**0.9021**	**0.9234**	**0.9113**
	MM-GANN-DDI	0.9066	0.9576	0.9989	0.8562	0.9150	0.8780
	MCFF-MTDDI	0.9010	0.9532	0.9984	0.8300	0.9122	0.8631
	MAEDDIE	0.8855	0.9426	0.9985	0.8120	0.9121	0.8464
	MDF-SA-DDI	0.8725	0.9385	0.9979	0.7518	0.9198	0.8220
	RANEDDI	0.8611	0.9225	0.9872	0.8155	0.9084	0.8110
	DDIMDL	0.8401	0.8824	0.9892	0.7678	0.8580	0.7800
Task 2	RaGSECo	**0.6675**	**0.7018**	**0.9862**	**0.5645**	**0.5875**	**0.5656**
	MM-GANN-DDI	0.6505	0.6546	0.9682	0.5562	0.5147	0.5230
	MCFF-MTDDI	0.6422	0.6564	0.9601	0.5407	0.5087	0.5140
	MAEDDIE	0.6462	0.6533	0.9680	0.5500	0.4781	0.4845
	MDF-SA-DDI	0.6333	0.6486	0.9667	0.5317	0.4900	0.4785
	DDIMDL	0.6039	0.6159	0.9737	0.5080	0.4465	0.4470
Task 3	RaGSECo	**0.4775**	**0.4362**	**0.9578**	**0.2962**	**0.2775**	**0.2782**
	MM-GANN-DDI	0.4657	0.4186	0.9355	0.2896	0.2500	0.2555
	MCFF-MTDDI	0.4545	0.4136	0.9340	0.2857	0.2348	0.2427
	MAEDDIE	0.4606	0.4099	0.9398	0.2875	0.2386	0.2421
	MDF-SA-DDI	0.4530	0.4092	0.9358	0.2768	0.2336	0.2399
	DDIMDL	0.4111	0.3739	0.9514	0.2715	0.1691	0.1823

Herein, we select three representative metric scores (Accuracy, AUPR, and F1) to evaluate the prediction performance of RaGSECo and its variants. Fig. 6 illustrates the metric scores of RaGSECo and its six variants on Tasks 1, 2, and 3 of Dataset 1. These figures show that RaGSECo achieves higher metric scores than its variants, indicating the effectiveness of RaGSE learning and propagation, multimodal DP representation learning, and the co-contrastive learning mechanism. Comparing RaGSECo-R and RaGSECo-M, we can see that RaGSECo-M performs better on Task 1, while RaGSECo-R significantly outperforms RaGSECo-M on Tasks 2 and 3. This observation confirms that the test DDIs include new drugs in Tasks 2 and 3, which may impact the model performance. Nevertheless, RaGSE propagation can effectively mitigate this issue. Analyzing the results of RaGSECo-I, RaGSECo-S, and RaGSECo-E reveals that incorporating the drug's initial features, SMILES string, and RaGSEs can enhance data diversity and improve the model's performance. Finally, the performance of RaGSECo-C validates the effectiveness of the co-contrastive learning mechanism.

4.6. Comparison with other methods

4.6.1. Dataset 1

To assess the competitiveness of our RaGSECo, we compared it with several state-of-the-art DDI prediction methods, namely MM-GANN-DDI, MCFF-MTDDI, MAEDDIE, MDF-SA-DDI, RANEDDI, and DDIMDL. Table 2 presents the metric scores achieved by these methods across the three tasks of Dataset 1. The comparison results demonstrate that our RaGSECo outperforms the competitors in three tasks. In Task 1, although RANEDDI also considers relation-aware graph structure information to learn drug embedding and achieve outstanding prediction performance, our RaGSECo achieves better results than RANEDDI. Specifically, the improvements of RaGSECo over RANEDDI are 2.33%, 1.81%, 3.74%, 2.83%, and 3.33% in terms of ACC, AUPR, Precision, Recall, and F1, respectively. The reasons causing this phenomenon are manifold. On the one hand, RANEDDI ignores multiple biological attributes, such as targets, enzymes, and chemical substructures. This makes RANEDDI lack the ability to capture relationships beyond interactions between drugs. On the other hand, simply concatenating

Table 4
The drug names and event types of the confirmed DDIs.

Rank	Drug names	DDI event type
1	Zileuton and Aprepitant	The metabolism decrease
2	Etravirine and Sulfisoxazole	The metabolism decrease
3	Sulfamethoxazole and Aprepitant	The metabolism decrease
4	Doxepin and Aprepitant	The metabolism decrease
5	Ramelteon and Etravirine	The metabolism decrease
6	Mefenamic acid and Estazolam	The excretion rate decrease
7	Lornoxicam and Clobazam	The excretion rate decrease
8	Cimicoxib and Clobazam	The excretion rate decrease
9	Antipyrine and Clofarabine	The excretion rate decrease
10	Palonosetron and Nisoldipine	The excretion rate decrease
11	Estazolam and Pipotiazine	The risk or severity of adverse effects increase
12	Flurbiprofen and Suprofen	The risk or severity of adverse effects increase
13	Naratriptan and Clotiazepam	The risk or severity of adverse effects increase
14	Zaleplon and Mometasone	The metabolism increase
15	Norgestrel and Primidone	The metabolism increase
16	Brigatiniband Dexamethasone	The metabolism increase
17	Hydroxyzine and Gemifloxacin	The risk or severity of QTc prolongation increase
18	Gemifloxacin and Cisapride	The risk or severity of QTc prolongation increase
19	Terbutaline and Gemifloxacin	The risk or severity of QTc prolongation increase
20	Lurasidone and Gemifloxacin	The risk or severity of QTc prolongation increase

Table 5
The hyper-parameters of best accuracy for the proposed RaGSECo on all Tasks.

Dataset	Task	bs	lr	dr	te	d'	n	d^{FNN}	t_{pos}	t_{neg}	λ
Dataset 1	Task 1	512	2e−5	0.3	120	500	0	1000	0.95	0.1	5
	Task 2	512	5e−6	0.2	120	500	3	1500	0.95	0.1	5
	Task 3	512	5e−6	0.2	120	500	3	1500	0.95	0.1	5
Dataset 2	Task 1	1024	2e−5	0.5	120	500	0	1500	0.95	0.1	5
	Task 2	1024	5e−6	0.5	120	500	3	1500	0.95	0.1	5
	Task 3	1024	5e−6	0.5	120	500	3	1500	0.95	0.1	5

embeddings of two drugs to obtain DP features also limits the generalization ability of RANEDDI. Both MAEDDIE and MDF-SA-DDI employ multiple biological attributes and consider drug similarities to learn drug embeddings, while our RaGSECo still performs better. The main reason is that neither MDF-SA-DDI nor MAEDDIE takes advantage of multi-relational interaction information between drugs.

For further insight, we investigate the performances of our RaGSECo and these six competitive baselines for each event. Fig. 7 displays the AUPR and AUC scores of the seven prediction models for each event on Task 1. As observed, RaGSECo produces greater AUPR and AUC scores than other methods in most event types. In most cases, RaGSECo can achieve a satisfactory result. All unsatisfactory prediction results are observed in low-frequency event types, such as #39, #52, and #64, with frequencies of only 98, 24, and 10 samples, respectively. The limited availability of training samples for these low-frequency event types may contribute to relatively poorer performance.

Tasks 2 and 3 compare our RaGSECo with five competitive prediction methods. RANEDDI only focuses on the embedding of known drugs, so RANEDDI cannot be applied to Tasks 2 and 3. In Tasks 2 and 3, the test DDIs contain new drugs, and the lack of interaction information on the new drugs weakens the generalization ability of models. Therefore, the prediction accuracy of models on Tasks 2 and 3 is lower than that on Task 1. Nevertheless, RaGSECo outperforms other competitors. MM-GANN-DDI incorporates multiple biological attributes and a multi-relational DDI graph to learn drug features. However, our RaGSECo model demonstrates superior performance. Specifically, RaGSECo outperforms MM-GANN-DDI by margins of 1.5%, 2.6%, 2.73%, 4.75%, and 2.8% in terms of ACC, AUPR, AUC, Recall, and F1, respectively. This improvement can be attributed to several factors. Firstly, MM-GANN-DDI overlooks the implicit correlations between DPs. Additionally, it fails to capture effective relation-aware graph structure information for new drugs.

4.6.2. Dataset 2

Table 3 presents the performance metrics of the proposed RaGSECo and several outstanding baseline approaches on three tasks of Dataset 2. Dataset 2 exhibits greater diversity compared to Dataset 1. It has more drugs, more DDIs, and a wider range of DDI event types. As observed, our RaGSECo can achieve better prediction performances than other competitors. In Task 1, compared with MCFF-MTDDI, the proposed RaGSECo has 3.34%, 2.73%, 7.21%, 1.12% and 4.82% performance improvements in terms of ACC, AUPR, Precision, Recall, and F1, respectively. As observed in Tasks 2 and 3, our RaGSECo still acquires the best results compared with five competitive prediction models. The experiment results demonstrate the effectiveness and robustness of RaGSECo. In conclusion, the reasons for RaGSECo outperforming are three folds: (1) RaGSECo enables all drugs, including new drugs, to capture effective relation-aware interaction information. (2) RaGSECo inventively combines initial features, SMILES information, and drug RaGSEs, enhancing the information diversity. (3) RaGSECo captures the underlying correlation between DPs and enables two views to supervise each other by co-contrastive learning.

4.7. Case study

In this section, we perform case studies to assess the effectiveness of RaGSECo. For these studies, we utilize Dataset 2, consisting of 1000 drugs. From this dataset, 800 drugs are randomly selected as known drugs, while the remaining 200 are categorized as new drugs. The 800 known drugs comprise 121,369 known DDIs. In contrast, the 200 new drugs contribute to $(200 * 200 - 200)/2 = 19{,}900$ DPs. Consequently, we train the RaGSECo model using the 121,369 known DDIs, along with their associated event types. We employ the 19,900 DPs for testing purposes.

The top-ranked prediction results are then reported. We pay attention to five events with the highest frequencies, and we check up the top 10 predictions related to each event. We test the selected DPs using the DDI Checker tool.[2] Out of the 50 selected DPs, 20 DDIs are confirmed and detailed in Table 4. For instance, the metabolism of Zileuton can be decreased when combined with Aprepitant. The risk or severity of QTc prolongation can be increased when Lurasidone is combined with Gemifloxacin.

5. Conclusion

In multi-relational DDI prediction, relation-aware graph embedding-based methods hold considerable promise. Nevertheless, interaction information of new drugs is unknown, which may cause these methods to suffer from severe over-fitting when predicting DDIs involving new drugs. To address this issue, we introduce the RaGSECo approach. The primary contribution of the RaGSECo is enabling all drugs to capture effective relation-aware interaction features and promoting the identical distribution of the training and test data. Additionally, our RaGSECo employs a cross-view contrastive mechanism to boost DP representation learning by utilizing the underlying correlation between DPs and supervising two views collaboratively. Our approach offers a promising drug and DP representation learning solution, thereby enhancing DDI prediction performance.

CRediT authorship contribution statement

Mengying Jiang: Formal analysis, Methodology, Writing – original draft. **Guizhong Liu:** Methodology, Writing – review & editing. **Biao Zhao:** Conceptualization, Methodology, Writing – review & editing. **Yuanchao Su:** Conceptualization, Methodology, Writing – review & editing. **Weiqiang Jin:** Methodology, Resources, Writing – review & editing.

[2] https://go.drugbank.com/interax/multi_search

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The URL link of the code for this paper is: https://github.com/mengyingjiang/RaGSECo.

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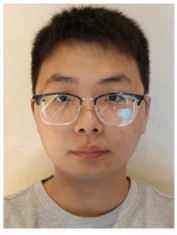
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